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Multi-Gait Learning for Humanoid Robots Using Reinforcement Learning with Selective Adversarial Motion Prior

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***Abstract*— Learning diverse locomotion skills for humanoid robots in a unified reinforcement learning framework remains challenging due to the conflicting requirements of stability and dynamic expressiveness across different gaits. We present a multi-gait learning approach that enables a humanoid robot to master five distinct gaits—walking, goose-stepping, running, stair climbing, and jumping—using a consistent policy structure, action space, and reward formulation. The key contribution is a selective Adversarial Motion Prior (AMP) strategy: AMP is applied to periodic, stability-critical gaits (walking, goose-stepping, stair climbing) where it accelerates convergence and suppresses erratic behavior, while being deliberately omitted for highly dynamic gaits (running, jumping) where its regularization would over-constrain the motion. Policies are trained via PPO with domain randomization in simulation and deployed on a physical 12-DOF humanoid robot through zero-shot sim-to-real transfer. Quantitative comparisons demonstrate that selective AMP outperforms a uniform AMP policy across all five gaits, achieving faster convergence, lower tracking error, and higher success rates on stability-focused gaits without sacrificing the agility required for dynamic ones.**

I. INTRODUCTION

Humanoid robots hold the promise of operating in environments designed for humans, but realizing this potential requires the ability to execute multiple locomotion behaviors adaptively [1]. Practical deployment demands not only walking for efficient traversal but also running for rapid movement, jumping for gap crossing or obstacle negotiation, stair climbing for multi-level navigation, and specialized gaits such as goose-stepping for structured or ceremonial scenarios. Each of these gaits imposes fundamentally different requirements on the control system: walking emphasizes energy efficiency and steady balance, running requires rapid force generation and aerial phase management, jumping demands maximal vertical impulse in a short time window, stair climbing needs precise foot placement and clearance control, and goose-stepping requires sustained leg extension with strict posture constraints. The central challenge is not merely acquiring each gait in isolation but learning them within a unified framework that shares policy architecture, observation space, and reward structure.

Reinforcement learning has emerged as a compelling approach for learning complex motor skills without requiring explicit analytical models. By interacting with a physics simulator at scale, RL agents can discover control strategies that are difficult to hand-design, especially for underactuated, high-degree-of-freedom systems like humanoid robots [2]. The key advantage of RL over traditional model-based control is its ability to handle nonlinear dynamics, contact switching, and actuator limits in a unified optimization process. Recent work has demonstrated impressive single-gait capabilities, including robust walking on flat and uneven ground, stair traversal with foot clearance, and basic disturbance recovery [3], [4]. However, these studies typically train a separate policy for each task, and each policy is tuned with its own reward design and hyperparameter configuration. Extending RL to multiple gaits under a single framework introduces new difficulties: the policy must reconcile conflicting dynamics across behaviors, the reward function must remain effective across gaits with vastly different motion characteristics, and the observation space must encode sufficient information for all behaviors without becoming redundant or misleading for any individual gait [5].

Adversarial Motion Prior (AMP) addresses a key difficulty in RL-based locomotion—producing natural, stable motions that transfer well to hardware [6]. By training a discriminator to distinguish between agent-generated and reference motion clips, AMP regularizes the policy toward demonstrated human-like behavior distributions. This has been shown to reduce joint jitter, improve foot placement consistency, and produce smoother trajectories that are more compatible with real actuator dynamics [7], [8]. However, existing work that employs AMP does so uniformly: once an AMP module is introduced, it remains active for all behaviors throughout training. This raises an important but underexplored question: *is AMP always beneficial, or can it be counterproductive for certain gait types?*

Our observation is that the effectiveness of AMP depends critically on the nature of the target gait. AMP excels at reinforcing periodic, repetitive motion patterns with moderate joint amplitude—exactly the characteristics of walking, goose-stepping, and stair climbing. For these gaits, the reference motion distribution closely matches the desired policy output, and the adversarial reward provides a stabilizing inductive bias that accelerates convergence and suppresses erratic behavior. Conversely, for gaits requiring large joint excursions, rapid acceleration, and explosive dynamics—such as running and jumping—the available human motion references are limited, and the AMP discriminator's regularization can act as an undesired constraint. In these cases,

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用于人形机器人的多步态学习——基于选择性对抗运动先验的强化学习

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摘要—在统一的强化学习框架中为 ***humanoid*** 机器人学习多样化的运动技能仍然具有挑战性，因为不同步态的稳定性和动态表现力要求相互冲突。我们提出了一种多步态学习方法，使 ***humanoid*** 机器人能够掌握五种不同的步态——行走、正步、奔跑、爬楼梯和跳跃——使用一致的策略结构、动作空间和奖励公式。关键贡献是一种选择性对抗运动先验（***AMP***）策略：***AMP*** 应用于周期性、稳定性关键的步态（行走、正步、爬楼梯），以加速收敛并抑制异常行为，而在高度动态的步态（奔跑、跳跃）中则故意省略，因为其正则化会过度约束运动。策略通过 ***PPO*** 在模拟中进行域随机化训练，并通过零样本模拟到现实的迁移部署在物理 ***12*** 自由度 ***humanoid*** 机器人上。定量比较表明，选择性 ***AMP*** 在所有五种步态中均优于均匀 ***AMP*** 策略，实现了更快的收敛速度、更低的跟踪误差，并在稳定性关键步态上获得了更高的成功率，同时未牺牲动态步态所需的敏捷性。

I. 引言

人形机器人有望在为人类设计的环境中运行，但要实现这一潜力需要能够自适应地执行多种运动行为 [1]。实际部署不仅要求行走以实现高效移动，还需要奔跑以实现快速移动，跳跃以跨越间隙或绕过障碍，爬楼梯以实现多级导航，以及在结构化或仪式场景中使用鹅步等专用步态。每种步态都对控制系统提出了根本不同的要求：行走强调能量效率和稳定平衡，奔跑需要快速产生力和空中阶段管理，跳跃要求在短时间内产生最大垂直冲量，爬楼梯需要精确的脚部放置和间隙控制，而鹅步要求持续腿部伸展并保持严格的姿态约束。核心挑战不仅在于孤立地获取每种步态，更在于在一个共享策略架构、观察空间和奖励结构的统一框架内学习它们。

强化学习已成为一种引人入胜的方法，用于学习复杂的运动技能，而无需显式的分析模型。通过与物理强化学习已成为一种引人入胜的方法，用于学习复杂的运动技能，而无需显式的分析模型。通过与大规模物理模拟器交互，RL智能体可以发现难以手动设计的控制策略，特别是对于欠驱动、高自由度系统，如人形机器人 [2]。RL相对于传统基于模型的控制的关键优势在于其能够在一个统一的优化过程中处理非线性动力学、接触切换和执行器限制。近期的研究展示了令人印象深刻的单步态能力，包括在平坦和不平坦地面上的稳健行走、带脚部间隙的楼梯跨越以及基本的干扰恢复 [3]，[4]。然而，这些研究通常为每个任务训练一个单独的策略，每个策略都使用其自己的奖励设计和超参数配置进行微调。将RL扩展到单一框架下的多个步态引入了新的困难：策略必须在行为之间协调冲突的动力学，奖励函数必须在具有截然不同运动特性的步态之间保持有效，观察空间必须为所有行为编码足够的信息，而不会对任何单个步态产生冗余或误导 [5]。

对抗运动先验 (AMP) 解决了基于强化学习 (RL) 的运动控制中的一个关键难题——生成自然、稳定的运动，并能很好地迁移到硬件 [6]。通过训练一个判别器来区分智能体生成的运动片段和参考运动片段，AMP 使策略规范化，使其更接近已展示的人类行为分布。研究表明，这可以减少关节抖动，提高脚部放置的一致性，并产生更平滑的轨迹，这些轨迹与真实执行器动力学更兼容 [7], [8]。然而，现有采用 AMP 的工作都是统一应用的：一旦引入 AMP 模块，它在整个训练过程中对所有行为都保持激活状态。这提出了一个重要但尚未深入探讨的问题：AMP 是否总是有益的，还是对某些步态类型可能适得其反？

我们的观察是，AMP 的有效性关键取决于目标步态的性质。AMP 在强化具有中等关节幅度的周期性、重复性运动模式方面表现出色——这正是行走、齐步走和爬楼梯的特征。对于这些步态，参考运动分布与期望的策略输出紧密匹配，对抗性奖励提供了一种稳定性的归纳偏差，从而加速收敛并抑制异常行为。相反，对于需要大关节运动、快速加减速和爆发性动力学的步态——例如跑步和跳跃——可用的人类运动参考有限，而 AMP 判别器的正则化可能成为不希望的约束。在这些情况下，

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the policy needs to explore beyond the reference manifold to discover strategies that fully utilize the robot's actuator capacity, and the AMP penalty can prevent the agent from reaching the necessary motion extremes.

This mismatch motivates a selective approach: rather than applying AMP universally or discarding it entirely, we propose to use it where it helps and disable it where it hinders. This strategy requires understanding which gait characteristics align with the AMP assumption and which deviate from it, and making an informed decision for each locomotion mode.

The contributions of this paper are as follows:

1) **A unified RL framework** that learns five practical gaits for humanoid robots—walking, goose-stepping, running, jumping, and stair climbing—using a consistent policy architecture and reward function based on sinusoidal joint reference tracking. All gaits share the same observation space, network structure, and reward formulation, differing only in reference trajectory parameters and reward weights.
2) **A selective Adversarial Motion Prior strategy** that applies AMP to periodic, stability-critical gaits while omitting it for highly dynamic, agile gaits. We provide empirical analysis demonstrating that AMP improves convergence speed and tracking accuracy for walking-type gaits but restricts motion amplitude and expressiveness for running and jumping.
3) **Full real-robot validation** of all five gaits with quantitative comparisons including tracking error, convergence speed, success/fall rate, and AMP versus non-AMP performance. We deploy every learned policy on a physical humanoid platform, providing a reproducible real-robot baseline for multi-gait learning.

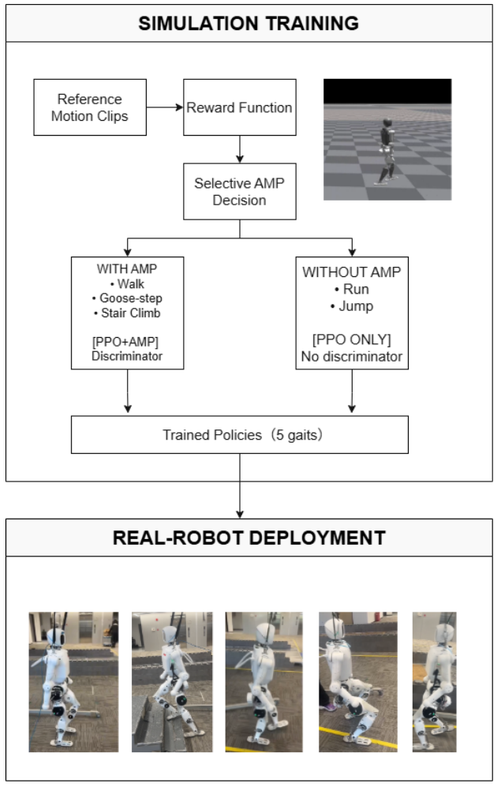


Fig. 1. Overview of the multi-gait learning pipeline. Policies are trained in simulation with or without AMP depending on gait type, then deployed on the physical robot.

II. Methodology

A. Overview

We propose a unified reinforcement learning framework that learns five distinct locomotion gaits for a 12-DOF humanoid robot: walking, goose-stepping, running, jumping, and stair climbing. The framework consists of three main components. First, a unified RL formulation with a consistent observation space, action space, and reward structure is used to train all five gaits, where each gait is distinguished only by its sinusoidal reference trajectory parameters and reward weight configuration. Second, a selective AMP strategy is applied: AMP is enabled for periodic, stability-critical gaits (walking, goose-stepping, stair climbing) and disabled for highly dynamic gaits (running, jumping). Third, policies trained in simulation with extensive domain randomization are deployed directly on a physical humanoid robot without fine-tuning. Figure 1 illustrates the overall pipeline.

B. Reinforcement Learning Formulation

We formulate each gait learning task as a Markov Decision Process (MDP) defined by the tuple $(\mathcal{S}, \mathcal{A}, r, \gamma, \rho_0)$. All five gaits share the same MDP structure, differing only in the reference trajectory generation, reward weight parameters, and AMP configuration.

1) State Space: The observation space received by the policy at each timestep t consists of proprioceptive states, reference signals, and command inputs. The per-frame observation vector $o_t \in \mathbb{R}^{50}$ is constructed as the concatenation of the following components:

- **Base linear velocity** $v_b \in \mathbb{R}^3$: expressed in the robot's body frame, scaled by a factor of 2.0.
- **Velocity commands** $c \in \mathbb{R}^3$: target linear velocity in x and y directions and angular velocity about the z axis.
- **Phase encoding** $\phi \in \mathbb{R}^2$: sinusoidal encoding $[\sin(2\pi\phi_t), \cos(2\pi\phi_t)]$ of the current gait phase.
- **Base angular velocity** $\omega_b \in \mathbb{R}^3$: expressed in the body frame, scaled by 0.25.
- **Projected gravity vector** $g_p \in \mathbb{R}^3$: the world-frame gravity vector $[0, 0, -1]^\top$ rotated by the base orientation.
- **Joint position error** $q_e \in \mathbb{R}^{12}$: the difference between current joint positions and the default standing configuration.

策略需要探索参考流形之外，以发现充分利用机器人执行器能力的策略，而 AMP 惩罚可能阻止智能体达到必要的运动极限。

这种不匹配促使我们采取一种选择性方法：既不普遍应用AMP，也不完全抛弃它，我们提出在它能发挥作用的地方使用它，在它造成阻碍的地方禁用它。这种策略需要理解哪些步态特征与AMP假设一致，哪些与之偏离，并对每种运动模式做出明智的决策。

本文的贡献如下：

1) 一个统一的RL框架，用于学习人形机器人的五种实用步态——行走、齐步、奔跑、跳跃和爬楼梯——该框架基于正弦关节参考跟踪的统一策略架构和奖励函数。所有步态共享相同的观测空间、网络结构和奖励公式，仅在参考轨迹参数和奖励权重上有所不同。

2) 一种选择性对抗运动先验策略，将AMP应用于周期性、稳定性关键的步态，而忽略高动态、敏捷的步态。我们提供了实证分析，证明AMP提高了行走类步态的收敛速度和跟踪精度，但限制了奔跑和跳跃的运动幅度和表现力。

3) 对所有五种步态进行完整的真实机器人验证，包括跟踪误差、收敛速度、成功/跌倒率和AMP与非AMP性能的定量比较。我们将每个学习到的策略部署在物理人形机器人平台上，为多步态学习提供可重复的真实机器人基准。

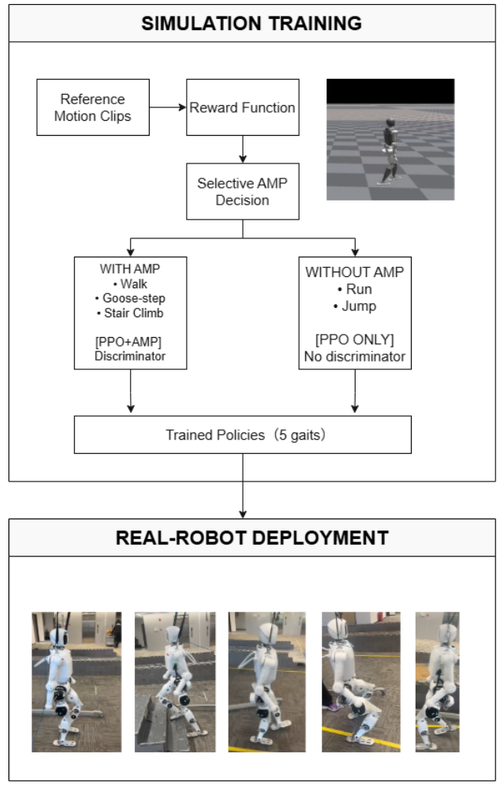


图1. 多步态学习流程概述。策略根据步态类型在模拟环境中使用或未使用AMP进行训练，然后部署到物理机器人上。

II. 方法论

A. 概述

我们提出了一种统一的强化学习框架，用于为12自由度人形机器人学习五种不同的运动步态：行走、齐步、跑步、跳跃和爬楼梯。该框架由三个主要组件组成。首先，使用具有一致观察空间、动作空间和奖励结构的统一RL公式来训练所有五种步态，其中每种步态仅通过其正弦参考轨迹参数和奖励权重配置来区分。其次，应用选择性AMP策略：AMP对周期性、稳定性关键的步态（行走、齐步、爬楼梯）启用，对高度动态的步态（跑步、跳跃）禁用。第三，在模拟中经过广泛领域随机化训练的策略直接部署在物理人形机器人上，无需微调。图1说明了整体流程。

B. 强化学习公式化

我们将每个步态学习任务表述为一个由元组$(\mathcal{S}, \mathcal{A}, r, \gamma, \rho_0)$定义的马尔可夫决策过程（MDP）。所有五种步态共享相同的MDP结构，仅在参考轨迹生成、奖励权重参数和AMP配置上有所不同。

参考轨迹生成、奖励权重参数和AMP配置。

1) 状态空间：策略在每个时间步接收的观测空间 t 由本体感觉状态、参考信号和指令输入组成。每帧观测向量 $o_t \in \mathbb{R}^{50}$ 由以下组件的连接构成：

- 基础线性速度 $v_b \in \mathbb{R}^3$：以机器人的本体坐标系表示，缩放因子为2.0。
- 速度指令 $c \in \mathbb{R}^3$：在 x和 y 方向上的目标线性速度以及绕 z 轴的角速度。
- 相位编码 $\phi \in \mathbb{R}^2$：当前步态相位[的正弦编码$\sin(2\pi\phi_t)$、余弦编码$\cos(2\pi\phi_t)$]。
- 基础角速度 $\omega_b \in \mathbb{R}^3$：在机体坐标系中表示，并乘以0.25。
- 投影重力向量 $g_p \in \mathbb{R}^3$：世界坐标系的重力向量 $[0, 0, -1]^\top$ 经过基础朝向旋转后的结果。
- 关节位置误差 $q_e \in \mathbb{R}^{12}$：当前关节位置与默认站立配置之间的差异。

- **Joint velocities** $\dot{q} \in \mathbb{R}^{12}$: scaled by 0.05.
- **Previous action** $a_{t-1} \in \mathbb{R}^{12}$: the action executed at the previous timestep.

The phase variable $\phi_t \in [0, 1)$ is advanced as $\phi_{t+1} = (\phi_t + \Delta t / T_{\text{cycle}}) \bmod 1$, where T_{cycle} is the gait-specific cycle time.

The reference joint trajectory is encoded through the phase variable. For periodic gaits (walking, goose-stepping, running, stair climbing), the reference is sinusoidal:

$$s = \sin(2\pi\phi_t) \tag{1}$$

$$q_{\text{ref}}^{\text{L,hip}} = -s \cdot \mathbb{1}_{s<0} \tag{2}$$

$$q_{\text{ref}}^{\text{L,knee}} = 2s \cdot \sigma \cdot \mathbb{1}_{s<0} \tag{3}$$

where σ is the target joint position scale (0.26 for walking and running, 0.28 for goose-stepping), and the right leg mirrors the left with opposite sign when $s > 0$. Each gait modulates this pattern through its $(T_{\text{cycle}}, \text{stance_ratio})$ pair: walking (0.8 s, 0.6), goose-stepping (0.7 s, 0.45), running (0.4 s, 0.35), stair climbing (0.7 s, 0.6).

For the jumping gait, a piecewise 4.0 s cycle trajectory is used instead, with five phases: squat ($\phi < 0.30$, $\sin^2$ knee flexion), takeoff (0.30–0.42, rapid extension), flight (0.42–0.48, hold extension), landing (0.48–0.75, impact absorption and recovery), and stand ($\phi \geq 0.75$, default pose). A curriculum progressively increases squat depth over 2000–10000 PPO iterations.

To capture temporal dynamics, the policy receives a history of $N = 21$ consecutive observation frames, stacked into a single input of dimension $50 \times 21 = 1050$. This enables implicit estimation of velocities and accelerations without explicit derivative computation.

The critic receives an extended observation s_t that additionally includes privileged simulation-only information: base linear and angular velocity, projected gravity, commands, joint position error, joint velocities, actions, phase encoding, friction coefficients, foot contact masks, stance masks, joint tracking error, push forces, foot vertical velocities, and foot clearance. This privileged vector (73 dimensions per frame) is stacked over 5 frames and concatenated with terrain height measurements (187 points), yielding 552 total dimensions.

2) Action Space: The policy outputs a 12-dimensional action vector $a_t \in \mathbb{R}^{12}$, corresponding to target joint position offsets for the 12 actuated DOFs (6 per leg: hip pitch, hip roll, hip yaw, knee, ankle pitch, ankle roll). The target joint position is computed as:

$$q_{\text{target}} = a_t \cdot k_{\text{action}} \cdot k_{\text{gain}} + q_{\text{default}} \tag{4}$$

where k_{action} is the action scale factor (0.25 for walking/stair climbing, 0.3 for running, 0.4 for goose-stepping) and $k_{\text{gain}} = \tau_{\max}/K_P$ is a per-joint gain that normalizes the action output relative to the joint's stiffness and torque limit.

The target positions are sent to a PD controller:

$$\tau = K_P(q_{\text{target}} - q) - K_D\dot{q} \tag{5}$$

The proportional gains K_P range from 75 to 280 N·m/rad for walking-type gaits (hip pitch and knee at 280, hip roll at 240, hip yaw at 75, ankle joints at 75), with derivative gains K_D from 5 to 14 N·m·s/rad. Goose-stepping employs significantly higher stiffness (hip pitch 450, knee 800, hip roll 300, hip yaw 150, ankle joints 200) and damping (K_D up to 40 on the knee), while running uses moderate increases (hip pitch and knee at 300, hip roll 240, hip yaw 100, ankle joints 80). Computed torques are clipped to per-joint maximum limits (60–150 N·m) specified in the URDF effort field. The policy executes at 50 Hz while the PD controller runs at 200 Hz (decimation factor of 4). An exponential moving average filter smooths actions, and a random action delay of 5–10 timesteps simulates real-world latency.

3) Reward Function: All five gaits share a unified reward formulation. The total reward at each timestep is a weighted sum:

$$r_{\text{total}} = \sum_i w_i r_i \tag{6}$$

The reward terms are organized into four categories:

(1) Joint tracking reward. The primary objective tracks the reference trajectory. For periodic gaits, the tracking reward for hip pitch and knee joints is computed as a dual-exponential sum:

$$r_{\text{track}} = 0.15 \sum_{j\in\{\text{hip,knee}\}} \sum_{i\in\{\text{L,R}\}} \left(e^{-4e_{ij}^2} + e^{-20e_{ij}^2}\right) \tag{7}$$

where $e_{ij} = q_{ij} - q_{\text{ref},ij}$ is the joint error, with the knee term masked to apply only during swing phase. Walking and running use a general tracking weight of $w_{\text{track}} = 2.0$, goose-stepping uses 4.0 (with tighter $\sigma = 0.15$ vs. 0.2), and a separate knee-only reward adds $w = 3.0$ (walking) or 25.0 (goose-stepping). For jumping, a single tracking reward ($w = 20.0$) computes mean squared error across all 12 joints.

(2) Base balance and posture reward. Includes orientation reward ($w = 3$ for walking and running, 5.0 for goose-stepping, 15.0 for jumping), base height reward ($w = 2$), and velocity tracking for linear ($w = 5.0$ walking, 8.0 goose-stepping, 20.0 running, 0.0 jumping) and angular ($w = 3.5$) velocity.

(3) Motion smoothness reward. Action smoothness penalizes first- and second-order changes ($w = -0.01$ for walking and running, -0.001 for goose-stepping), joint velocity ($w = -10^{-4}$), and rated torque ($w = -0.02$ per side for walking, -0.03 for goose-stepping, -0.05 for running). Goose-stepping adds leg straightness ($w = 12.0$), calf lift ($w = 8.0$), and foot kick ($w = 6.0$) terms.

(4) Safety and gait-specific rewards. Fall termination when the base contacts ground or $|\text{roll}| > 0.8$ rad, $|\text{pitch}| > 1.0$ rad. Knee collision penalties are applied. Periodic gaits include feet swing height ($w = 2$) and alternate foot swing ($w = 6.0$ for goose-stepping). Jumping uses specialized rewards: jump height targeting ($w = 150.0$, Gaussian reward centered at 0.275 m with a standing-phase penalty to prevent spurious small hops), vertical takeoff velocity ($w = 120.0$, piecewise nonlinear: linear ramp below 1.0 m/s, quadratic

- 关节速度 $\dot{q} \in \mathbb{R}^{12}$：已乘以 ***0.05***。
- 前一步动作 $a_{t-1} \in \mathbb{R}^{12}$：在先前时间步执行的动作。

相位变量 $\phi_t \in [0, 1)$ 按照公式 $\phi_{t+1} = (\phi_t + \Delta t / T_{\text{cycle}}) \bmod 1$ 进行更新，其中 T_{cycle} 是特定步态的周期时间。

参考关节轨迹通过相位变量进行编码。对于周期性步态（行走、齐步、跑步、爬楼梯），参考轨迹是正弦波：

$$s = \sin(2\pi\phi_t) \tag{1}$$

$$q_{\text{ref}}^{\text{L,hip}} = -s \cdot \mathbb{1}_{s<0} \tag{2}$$

$$q_{\text{ref}}^{\text{L,knee}} = 2s \cdot \sigma \cdot \mathbb{1}_{s<0} \tag{3}$$

其中 σ 是目标关节位置缩放系数（行走和跑步为 0.26，齐步为 0.28），当 $s > 0$ 时，右腿以相反符号镜像左腿。每种步态通过其 $(T_{\text{cycle}}$，支撑比) 对来调整该模式：行走（0.8 秒，0.6）、齐步（0.7 秒，0.45）、跑步（0.4 秒，0.35）、爬楼梯（0.7 秒，0.6）。

对于跳跃步态，采用一个分段4.0秒周期的轨迹，包含五个阶段：下蹲（$\phi < 0.30$，正弦^2 屈膝）、起跳（0.30–0.42，快速伸展）、飞行（0.42– 0.48，保持伸展）、着陆（0.48–0.75，冲击吸收和恢复）和站立（$\phi \geq 0.75$，默认姿态）。课程逐渐增加下蹲深度，跨越2000– 10000 PPO迭代次数。

为了捕捉时序动态，策略接收一个由 $N = 21$ 连续观测帧组成的记录，堆叠成一个维度为 $50 \times 21 = 1050$ 的单一输入。这允许隐式估计速度和加速度，而无需显式计算导数。

评价者接收一个扩展观测 s_t，其中额外包含特权模拟信息：基础线速度和角速度、投影重力、指令、关节位置误差、关节速度、动作、阶段编码、摩擦系数、脚部接触掩码、站立掩码、关节跟踪误差、推力、脚部垂直速度和脚部间隙。这个特权向量（每帧73维度）堆叠5帧并连接地形高度测量（187点），总共产生552个维度。

2) 动作空间：策略输出一个12维的动作向量 $a_t \in \mathbb{R}^{12}$，对应于12个主动DOF（每条腿6个：髋关节俯仰、髋关节滚动、髋关节偏航、膝关节、踝关节俯仰、踝关节滚动）的目标关节位置偏移。目标关节位置计算如下：

$$q_{\text{target}} = a_t \cdot k_{\text{action}} \cdot k_{\text{gain}} + q_{\text{default}} \tag{4}$$

其中 k_{action} 是动作缩放因子（行走/爬楼梯为0.25，跑步为0.3，行进为0.4），$k_{\text{gain}} = \tau_{\max}/K_P$ 是每个关节的增益，用于相对于关节的刚度和扭矩限制来归一化动作输出。

目标位置被发送到PD控制器：

$$\tau = K_P(q_{\text{target}} - q) - K_D\dot{q} \tag{5}$$

对于步行类型的步态（髋关节俯仰和膝关节为280，髋关节侧倾为240，髋关节偏航为75，踝关节为75），比例增益 K_P 的范围为75至280 N·m/rad，微分增益 K_D 的范围为5至14 N·m·s/rad。齐步走采用显著更高的刚度（髋关节俯仰450，膝关节800，髋关节侧倾300，髋关节偏航150，踝关节200）和阻尼（K_D 膝关节最高达40），而跑步则使用适度的增加（髋关节俯仰和膝关节为300，髋关节侧倾240，髋关节偏航100，踝关节80）。计算出的扭矩被裁剪到URDF力场中指定的每个关节最大限制（60–150 N·m）。策略以50 Hz执行，而PD控制器以200 Hz运行（降采样因子为4）。指数移动平均滤波器平滑动作，5–10步长的随机动作延迟模拟了现实世界的延迟。

3) 奖励函数：所有五种步态共享统一的奖励公式。每个时间步长的总奖励是一个加权求和：

$$r_{\text{total}} = \sum_i w_i r_i \tag{6}$$

奖励项分为四个类别：

(1) 关节跟踪奖励。主要目标跟踪参考轨迹。对于周期性步态，髋关节俯仰和膝关节的跟踪奖励计算为双指数和：

$$r_{\text{track}} = 0.15 \sum_{j\in\{\text{hip,knee}\}} \sum_{i\in\{\text{L,R}\}} \left(e^{-4e_{ij}^2} + e^{-20e_{ij}^2}\right) \tag{7}$$

在 $e_{ij} = q_{ij} - q_{\text{ref}}$ 和 $_{ij}$ 处，这是关节误差，其中膝盖项被屏蔽，仅在摆动阶段应用。行走和跑步使用通用跟踪权重 $w_{\text{track}} = 2.0$，踱步使用 4.0（更紧密的 $\sigma = 0.15$ 对比 0.2），并且添加了单独的膝盖奖励，为行走时增加 $w = 3.0$ 或为踱步增加 25.0。对于跳跃，单个跟踪奖励 ($w = 20.0$) 计算所有 12 个关节的均方误差。

(2) 基础平衡和姿态奖励。包括方向奖励（行走和跑步时为 $w = 3$，踱步时为 5.0，跳跃时为 15.0）、基础高度奖励 ($w = 2$)，以及线速度（行走时为 $w = 5.0$，踱步时为 8.0，跑步时为 20.0，跳跃时为 0.0）和角速度 ($w = 3.5$) 的跟踪。

(3) 运动平滑度奖励。动作平滑度惩罚一阶和二阶变化（$w = -0.01$ 用于行走和跑步，-0.001 用于踏步）、关节速度（$w = -10^{-4}$）和额定扭矩（$w = -0.02$ 每侧用于行走，-0.03 用于踏步，-0.05 用于跑步）。踏步会增加腿部挺直（$w = 12.0$）、小腿抬高（$w = 8.0$）和脚跟踢（$w = 6.0$）项。

(4) 安全性和步态特定奖励。当基座接触地面或 |侧倾| > 0.8 角度、 |俯仰| >1.0弧度时终止跌倒。应用膝盖碰撞惩罚。周期性步态包括脚部摆动高度($w = 2$)和交替脚部摆动($w = 6.0$ 用于踏步)。跳跃使用专用奖励：跳跃高度目标($w = 150.0$，以0.275米为中心的高斯奖励，站立阶段惩罚以防止虚假的小跳跃），垂直起跳速度($w = 120.0$，分段非线性：1.0米/秒以下为线性斜坡，超过部分为二次

above), and synchronized feet contact ($w = 20.0$), while penalizing horizontal velocity ($w = -15.0$).

The key design principle is that *all five gaits use the same reward formulation*—diversity arises from reference specification, weight tuning, and gait-specific terms rather than structural differences.

C. Selective AMP Strategy

1) Brief Review of AMP: Adversarial Motion Prior (AMP) combines adversarial imitation learning with reinforcement learning to produce motions that resemble a set of reference demonstrations. The method introduces a discriminator network D_ϕ trained to distinguish between state transitions from expert motion clips and state transitions produced by the policy during execution.

The discriminator architecture is a 3-layer MLP with hidden dimensions $[1024, 512, 256]$ and ReLU activations. The input to the discriminator is the concatenation of the current state s_t and the next state s_{t+1}, capturing a single state transition. The discriminator outputs a scalar logit $D(s_t, s_{t+1})$, which is converted into a reward signal:

$$r_{\text{AMP}} = \alpha \cdot \max\left(0, 1 - \frac{1}{4}(D(s_t, s_{t+1}) - 1)^2\right) \tag{8}$$

where α is a per-gait coefficient scaling the AMP reward magnitude. This reward is then linearly interpolated with the task reward:

$$r_{\text{combined}} = (1 - \beta) \cdot r_{\text{AMP}} + \beta \cdot r_{\text{task}} \tag{9}$$

where $\beta \in [0, 1]$ controls the balance between AMP regularization and task optimization.

The discriminator is trained with a binary cross-entropy loss combined with a gradient penalty for stability:

$$\mathcal{L}_D = \mathbb{E}_{\text{policy}}[(D+1)^2] + \mathbb{E}_{\text{expert}}[(D-1)^2] + \lambda \cdot \mathbb{E}_{\text{expert}}[\|\nabla_{(s,s')} D\|^2] \tag{10}$$

where the gradient penalty coefficient $\lambda = 10$. The reference motion data consists of captured human locomotion sequences, processed into 200,000 preloaded state transitions.

2) Our Selection Criterion: The central insight of our approach is that AMP's regularization effect is beneficial for some gaits but detrimental for others. We propose a per-gait selection criterion based on motion characteristics:

AMP is applied to: Walking, goose-stepping, and stair climbing. These gaits are periodic with well-defined, repeating motion cycles and moderate joint amplitudes, where stability is the primary concern. For these gaits, the reference motion distribution closely matches the desired policy output, and the adversarial reward provides a stabilizing inductive bias that accelerates convergence, suppresses erratic behavior, and produces smoother joint trajectories compatible with real actuator dynamics. The AMP coefficient α and interpolation β are tuned per gait: walking and stair climbing use $\alpha = 0.3$, $\beta = 0.8$; goose-stepping uses $\alpha = 0.6$, $\beta = 0.7$ (higher α to enforce the distinctive straight-leg style; lower β to allow more AMP influence).

AMP is NOT applied to: Running and jumping. These gaits involve large joint excursions that may exceed the range of available reference data, and require rapid acceleration and explosive dynamics. For running, setting $\alpha = 0.0$ and $\beta = 1.0$ removes AMP entirely, giving the policy full freedom to discover high-speed strategies using the actuator's complete torque capacity (up to 150 N·m on knee joints). Similarly, jumping uses $\alpha = 0.0$, $\beta = 1.0$, and disables the symmetry loss (sym_loss = False) to allow asymmetric force generation during takeoff. In both cases, the AMP discriminator would penalize the policy for deviating from the reference distribution, actively discouraging the motion extremes needed for sufficient speed or jump height.

In summary: *AMP improves stability but limits expressiveness; we select its use adaptively based on gait characteristics.* Table I summarizes the configuration for each gait.

TABLE I
SELECTIVE AMP CONFIGURATION PER GAIT.

Gait	α	β	Sym Loss
Walking	0.3	0.8	True
Goose-stepping	0.6	0.7	True
Running	0.0	1.0	True
Stair climbing	0.3	0.8	True
Jumping	0.0	1.0	False

D. Training Pipeline

1) Simulation Training: All policies are trained using the Proximal Policy Optimization (PPO) algorithm [9] in the Isaac Gym physics simulator [5] with 4096 parallel environments. The actor network uses an MLP with hidden layers [512, 256, 128] and ELU activation. The critic network shares the same architecture. The PPO algorithm runs 5 learning epochs over 4 mini-batches per iteration, with a learning rate of 5×10^{-4}, discount factor $\gamma = 0.99$, and GAE parameter $\lambda = 0.95$. The KL divergence target is set to 0.01 with an adaptive learning rate schedule. Training proceeds for up to 40,000 iterations, with the empirical observation normalizer updated online during training [10].

2) Domain Randomization: To bridge the simulation-to-reality gap, we apply extensive domain randomization during training. The following parameters are randomized at the environment level each episode:

- **Contact properties**: Static and dynamic friction uniformly sampled from $[0.2, 1.5]$; restitution from $[0.0, 1.0]$.
- **Base mass**: Perturbed by $[-2, +5]$ kg from the nominal mass.
- **Link inertia**: Scaled by $[0.5, 1.8]$ independently per link.
- **Center of mass**: Displaced by $[-0.1, 0.1]$ m in each axis.
- **Motor strength**: Randomized to $[0.7, 1.0]$ (walking/stair climbing), $[0.8, 1.2]$ (running/goose-stepping), or $[0.9, 1.1]$ (jumping).

函数），以及同步脚部接触($w = 20.0$)，同时惩罚水平速度($w = -15.0$)。

核心设计原则是：所有五种步态都使用相同的奖励公式——多样性源于参考规范、权重调优和步态特定项，而非结构差异。

C. 选择性AMP策略

1) AMP简要回顾：对抗运动先验（AMP）结合对抗模仿学习与强化学习，生成类似一组参考演示的运动。该方法引入了一个判别器网络 D_ϕ，用于区分专家运动片段的状态转换与策略执行期间产生的状态转换。

判别器架构是一个具有隐藏维度 $[1024, 512, 256]$ 和ReLU激活的三层MLP。判别器的输入是当前状态 s_t 与下一个状态 s_{t+1}的串联，捕捉单个状态转换。判别器输出一个标量logit$D(s_t, s_{t+1})$，该logit被转换为奖励信号：

$$r_{\text{AMP}} = \alpha \cdot \max\left(0, 1 - \frac{1}{4}(D(s_t, s_{t+1}) - 1)^2\right) \tag{8}$$

其中 α 是每个步态的系数，用于缩放AMP奖励幅度。此奖励随后与任务奖励线性插值：

$$r_{\text{combined}} = (1 - \beta) \cdot r_{\text{AMP}} + \beta \cdot r_{\text{task}} \tag{9}$$

$\beta \in [0, 1]$ 控制着AMP正则化和任务优化之间的平衡。

判别器使用二元交叉熵损失结合梯度惩罚进行训练以保持稳定性：

$$\mathcal{L}_D = \mathbb{E}_{\text{policy}}[(D+1)^2] + \mathbb{E}_{\text{expert}}[(D-1)^2] + \lambda \cdot \mathbb{E}_{\text{expert}}[\|\nabla_{(s,s')} D\|^2] \tag{10}$$

其中梯度惩罚系数为 $\lambda = 10$。参考运动数据由捕获的人类运动序列组成，处理为200,000个预加载的状态转换。

2) 我们的选取标准：我们方法的核心洞察是，AMP的正则化效果对某些步态有益，但对其他步态有害。我们基于运动特征提出了每步态的选取标准：

AMP 应用于：行走、正步和爬楼梯。这些步态具有周期性、定义明确且重复的运动周期以及适度的关节幅度，其中稳定性是主要关注点。对于这些步态，参考运动分布closely matchesthe desiredpolicy output，而对抗性奖励提供了一种稳定性的归纳偏置，可以加速收敛、抑制异常行为，并产生与实际执行器动力学兼容更平滑的关节轨迹。AMP 系数 α 和插值 β 按步态分别调整：行走和爬楼梯使用 $\alpha = 0.3$, $\beta = 0.8$；正步使用 $\alpha = 0.6$, $\beta = 0.7$ (更高的 α 来强化独特的直腿风格；更低的β 来允许更多AMP影响)。

AMP不应用于：跑步和跳跃。这些步态涉及大关节运动范围，可能超出可用参考数据的范围，并且需要快速加速和爆发动力学。对于跑步，设置 $\alpha = 0.0$ 和 $\beta = 1.0$将完全移除AMP，使策略能够自由地使用执行器的完整扭矩容量（膝关节为150 N·m）发现高速策略。类似地，跳跃使用 $\alpha = 0.0$、 $\beta = 1.0$，并禁用对称损失（sym loss = False）以允许在起跳时产生非对称力。在这两种情况下，AMP判别器会因策略偏离参考分布而进行惩罚，主动抑制实现足够速度或跳跃高度所需的运动极端情况。

总而言之：AMP 提高稳定性但限制了表现力；我们根据步态特征自适应地选择其使用。表 I 总结了每种步态的配置。

表 I 每种步态的Selective AMP配置。

Gait	α	β	Sym Loss
行走	0.3	0.8	True
鹅步	0.6	0.7	True
奔跑 g	0.0	1.0	True
爬楼梯	0.3	0.8	True
跳跃	0.0	1.0	False

D. 训练流程

1) 模拟训练：所有策略均使用近端策略优化（PPO）算法 [9] 在Isaac Gym物理模拟器 [5] 的4096个并行环境中进行训练。Actor网络使用具有隐藏层的MLP和ELU激活函数。Critic网络共享相同的架构。PPO算法在每个迭代中运行5个学习周期，每个周期处理4个小批量，学习率为 5×10^{-4}，折扣因子为 $\gamma = 0.99$，GAE参数为 $\lambda = 0.95$。KL散度目标设置为0.01，并采用自适应学习率计划。训练最多进行40,000次迭代，经验观察正常化器在训练过程中 [10]在线更新。

2) 域随机化：为了弥合模拟与现实之间的差距，我们在训练过程中应用广泛的域随机化。每个回合在环境级别随机化的参数如下：

- 接触属性：静摩擦力和动摩擦力从 $[0.2, 1.5]$ 均匀采样；恢复系数来自 $[0.0, 1.0]$。
- 基础质量：从标称质量扰动 $[-2, +5]$ kg。
- 连杆惯性：每个连杆独立缩放 $[0.5, 1.8]$ 。
- 质心：每个轴位移 $[-0.1, 0.1]$ m。
- 电机力量：随机分配至 $[0.7, 1.0]$ (行走/爬楼梯)、$[0.8, 1.2]$ (奔跑/小碎步) 或 $[0.9, 1.1]$ (跳跃)。

- **PD gains**: K_P scaled by $[0.8, 1.2]$, K_D scaled by $[0.8, 1.2]$.
- **Joint properties**: Joint friction $[0.3, 1.5]$, damping $[0.3, 4.0]$, armature $[0.8, 1.2]$.
- **Action lag**: Actions delayed by a random number of timesteps from $[5, 10]$, simulating control latency.
- **External disturbances**: Random impulses applied to the base with maximum linear velocity 2.0 m/s and angular velocity 1.5 rad/s, at intervals of 10 seconds.
- **Initial state**: Joint positions scaled by $[0.5, 1.5]$ and offset by $[-0.1, 0.1]$ rad at reset.

A curriculum is employed for action lag: the minimum delay is gradually increased from 2 to 5 timesteps over the course of training, starting from a milder setting to facilitate initial learning and progressively increasing the difficulty.

3) Sim-to-Real Transfer: Trained policies are deployed on the physical humanoid robot using zero-shot transfer—no fine-tuning on real hardware is performed. The deployment pipeline consists of:

1) **Policy export**: The trained actor network is exported to ONNX format with the empirical observation normalizer as a single inference module, and loaded via ONNX Runtime on the robot's onboard computer.
2) **Onboard execution**: The policy runs onboard the robot's computer at 50 Hz, receiving proprioceptive observations from an external IMU (angular velocity and orientation) and joint encoders (position and velocity). Since base linear velocity is not directly measurable on the hardware, the corresponding observation channels are set to zero, relying on the observation history stack to implicitly capture velocity information.
3) **PD control**: Joint torques are computed by the onboard PD controller at 500 Hz using the target positions output by the policy. A first-order low-pass filter with cutoff frequency 50 Hz smooths the policy actions before conversion to joint targets.

The domain randomization applied during training ensures that the policy is robust to the discrepancies between simulated and real dynamics, including differences in motor response, contact behavior, and sensor noise.

III. Experiments

The video of the experiments can be found at `https://linqi-ye.github.io/video/lite.mp4`

A. Experimental Setup

Experiments were conducted on the Lite_11 humanoid lower-body platform, which provides 12 actuated joints spanning the bilateral hip, knee, and ankle degrees of freedom. All policies were trained in Isaac Gym and then deployed on the physical robot for validation. We considered five locomotion modes: walking, goose-stepping, stair climbing, running, and jumping. Stair climbing includes both ascent and descent, whereas jumping consists of in-place two-foot takeoff followed by landing recovery. All tasks share the same policy interface: a 12-dimensional action vector, a 50-dimensional per-frame observation, and a 21-frame history stack, resulting in a 1050-dimensional policy input. Consequently, gait diversity is induced by reference trajectories, reward allocation, and motion-prior configuration rather than by changing the control architecture.

All tasks use a unified simulation and control backbone. The physics step is 0.005 s, and the policy is updated every four simulation steps, yielding an effective control period of 0.02 s. Low-level execution is handled by a PD position controller, which converts the policy-produced joint target offsets into actuation commands. PPO is used for all experiments. Except for jumping, which uses a slightly higher entropy coefficient to accommodate stronger exploration demands, the same base hyperparameters are shared across tasks. The main training settings are summarized in Table II.

Task environments and prior settings are defined according to gait characteristics. Walking, goose-stepping, and running are trained primarily on flat or low-complexity terrain, stair climbing is trained on triangle-mesh stairs with curriculum learning, and jumping is trained on flat terrain to emphasize vertical impulse generation, aerial stabilization, and landing recovery. Selective AMP is central to the experimental design: walking, goose-stepping, and stair climbing belong to the AMP group, whereas running and jumping are trained without AMP. This partition reflects the observation that periodic, stability-oriented motions benefit from motion priors, whereas highly dynamic motions depend more strongly on expressive freedom and task-driven optimization. To improve sim-to-real transfer robustness, domain randomization is applied throughout training, including friction, added mass, center-of-mass perturbation, motor strength, PD gains, inertia, joint friction, joint damping, initial joint scaling and bias, and action delay. Stair climbing uses a stronger terrain curriculum, while jumping adopts a more conservative randomization range.

B. Evaluation Metrics

To evaluate multi-gait policies from both optimization and execution perspectives, we consider joint tracking quality, convergence speed, task reliability, training stability, and posture stability. Performance is not judged solely by whether a motion emerges, but by whether the learned motion adheres to the intended structure, converges stably during training, and remains controllable during execution.

The joint tracking metric characterizes how well the policy preserves the intended kinematic organization of each gait. For walking, goose-stepping, and stair climbing, it primarily reflects coordination between stance and swing phases; for running and jumping, it reflects the ability to maintain the target motion pattern under high-speed or explosive dynamics. Convergence steps quantify the optimization cost required to reach a stable regime. Tasks that stabilize within fewer iterations are easier to optimize under the unified formulation, whereas tasks that require more iterations place stronger demands on contact switching, posture recovery, or temporal coordination.

At execution time, success rate and fall rate describe task reliability. A high success rate indicates that the robot can

- PD增益： K_P 按 $[0.8, 1.2]$比例缩放， K_D 按$[0.8, 1.2]$比例缩放。
- 关节属性：关节摩擦力 $[0.3, 1.5]$、阻尼$[0.3, 4.0]$、电枢$[0.8, 1.2]$。
- 动作延迟：动作随机延迟 $[5, 10]$，个时间步，模拟控制延迟。
- 外部干扰：以 10 秒为间隔，对基座施加最大线速度 2.0 m/s 和角速度 1.5 rad/s 的随机脉冲。
- 初始状态：关节位置缩放 $[0.5, 1.5]$ 并偏移 $[-0.1, 0.1]$ rad 在重置时。

对于动作延迟，采用课程学习：最小延迟从 2 个时间步逐渐增加到 5 个时间步，在训练过程中逐步增加难度，从较温和的设置开始以促进初始学习。

3) 模拟到现实迁移：训练好的策略通过零样本迁移部署在物理人形机器人上——不进行真实硬件上的微调。部署流程包括：

1) 策略导出：训练好的演员网络被导出为ONNX格式，并将经验观察归一化器作为单个推理模块，通过ONNX Runtime在机器人板载计算机上加载。2) 板载执行：策略以50 Hz的频率在机器人计算机上运行，接收来自外部IMU（角速度和方向）和关节编码器（位置和速度）的本体感觉观察。由于硬件上无法直接测量基座线性速度，相应的观察通道被设置为0，依靠观察历史栈隐式捕获速度信息。3) PD控制：板载PD控制器以500 Hz的频率计算关节力矩，使用策略输出的目标位置。一阶低通滤波器以50 Hz的截止频率平滑策略动作，然后再转换为关节目标。

训练期间应用的域随机化确保策略能够抵抗模拟和真实动力学之间的差异，包括电机响应、接触行为和传感器噪声的差异。

III. 实验

实验视频可在 https: // linqi-ye.github.io/video/lite.mp4 找到

A. 实验设置

在 Lite 11 人形机器人下肢平台上进行了实验，该平台提供12个驱动关节，覆盖双侧髋关节、膝关节和踝关节的自由度。所有策略均在 Isaac Gym 中进行训练，然后部署在物理机器人上进行验证。我们考虑了五种运动模式：行走、齐步走、爬楼梯、跑步和跳跃。爬楼梯包括上升和下降，而跳跃则包括原地双脚起跳和落地恢复。所有任务共享相同的策略接口：一个12维动作向量，一个50维的每帧观测值，以及一个21帧的历史记录堆叠，导致得到一个1050维的策略输入。因此，步态多样性是由参考轨迹、奖励分配和运动先验配置引入的，而不是通过改变控制架构。

所有任务使用统一的仿真和控制骨干。物理步长为 0.005 s，策略每四步仿真步更新一次，产生有效的控制周期为 0.02 s。低级执行由 PD 位置控制器处理，将策略产生的关节目标偏移量转换为驱动指令。所有实验均使用 PPO。除了跳跃使用略高的熵系数以适应更强的探索需求外，所有任务共享相同的基线超参数。主要的训练设置总结在表 II 中。

任务环境和先验设置根据步态特征定义。行走、齐步和跑步主要在平坦或低复杂度地形上训练，楼梯攀爬在三角形网格楼梯上使用课程学习进行训练，跳跃在平坦地形上进行以强调垂直冲量生成、空中稳定和着陆恢复。选择性 AMP 是实验设计的核心：行走、齐步和楼梯攀爬属于 AMP 组，而跑步和跳跃则不使用 AMP。这种划分反映了观察到的周期性、稳定性导向的运动受益于运动先验，而高度动态的运动则更依赖于表达自由和任务驱动的优化。为了提高仿真到现实的迁移鲁棒性，在整个训练过程中应用了域随机化，包括摩擦、附加质量、质心扰动、电机强度、PD 增益、惯性、关节摩擦、关节阻尼、初始关节缩放和偏差以及动作延迟。楼梯攀爬使用更强的地形课程，而跳跃采用更保守的随机化范围。

B. 评估指标

为了从优化和执行两个角度评估多步态策略，我们考虑了联合跟踪质量、收敛速度、任务可靠性、训练稳定性和位置稳定性。性能的评判不仅在于是否产生了运动，更在于学习到的运动是否遵循预期结构、在训练过程中是否稳定收敛，以及在实际执行中是否保持可控性。

联合跟踪指标描述了策略如何保持每种步态的预期运动学组织结构。对于行走、齐步和爬楼梯，它主要反映了支撑相和摆动相之间的协调性；对于跑步和跳跃，它则反映了在高速或爆发性动力学下维持目标运动模式的能力。收敛步数量化了达到稳定状态所需的优化成本。在统一框架下，能在较少迭代中稳定化的任务更容易优化，而需要更多迭代的任务则对接触切换、姿态恢复或时间协调提出了更高要求。

在执行时，成功率与跌倒率描述了任务可靠性。高成功率表明机器人能够

TABLE II

MAIN TRAINING CONFIGURATIONS FOR THE FIVE GAIT TASKS.

Gait	# Envs	Obs. Dim.	Act. Dim.	Sim Step	Control Period	num_steps_per_env	Max Iterations	AMP Group
Walking	4096	1050	12	0.005 s	0.02 s	24	40001	AMP
Goose-stepping	4096	1050	12	0.005 s	0.02 s	24	20001	AMP
Stair climbing	4096	1050	12	0.005 s	0.02 s	24	40001	AMP
Running	2048	1050	12	0.005 s	0.02 s	24	40001	No AMP
Jumping	2048	1050	12	0.005 s	0.02 s	24	11000	No AMP

repeatedly sustain the desired gait, whereas a high fall rate typically indicates sensitivity to initial conditions, rhythm control, or dynamic recovery. Training error characterizes the smoothness of optimization: lower error generally corresponds to more stable parameter updates, whereas persistently large error suggests stronger oscillation and higher uncertainty. Posture stability measures whole-body control quality. Higher posture stability corresponds to smaller torso oscillation, smoother roll-pitch regulation, and more consistent phase transitions. These metrics are used below to compare both overall gait performance and the distinction between AMP and non-AMP training.

C. Results and Analysis

1) Overall Performance of Five Gaits: Under a unified policy structure, shared observation design, and common reward organization, the proposed framework learns and deploys five locomotion modes: walking, goose-stepping, stair climbing, running, and jumping. These tasks span the spectrum from low-dynamic periodic locomotion to highly dynamic explosive motion, thereby testing the scalability of the unified RL formulation across substantially different gait classes.

Qualitatively, walking yields a regular forward stepping pattern; goose-stepping exhibits the intended high leg lift and straight-knee swing; stair climbing maintains sufficient foot clearance and body posture during both ascent and descent; running produces shorter stance durations and stronger forward propulsion; and jumping realizes the full motion chain from squat preparation and rapid takeoff to aerial stabilization and landing recovery. Figure 2 presents representative real-robot image sequences for the five gaits. These results indicate that the unified framework is able not only to maintain balance, but also to generate motion patterns with clearly differentiated dynamic characteristics and stylistic features.

Table III summarizes the quantitative performance of the five gaits. The reported indicators correspond to joint tracking quality, convergence speed, task reliability, training stability, and posture stability. Because the reward composition and task difficulty differ across gaits, these values are intended to characterize training and execution behavior under the unified framework rather than to serve as strictly normalized physical quantities across tasks.

2) Effect of AMP on Different Gaits: This subsection examines the operating boundary of AMP across gait classes. In general, AMP is most effective for gaits with clear periodic structure and stability-first objectives, whereas it can become restrictive for highly dynamic and strongly explosive motions. Walking, goose-stepping, and stair climbing fall into the former category: they require stable stance–swing transitions, consistent joint coordination, and low postural fluctuation. For such tasks, motion priors reduce ineffective exploration and typically yield lower tracking error, smoother reward growth, and faster convergence.

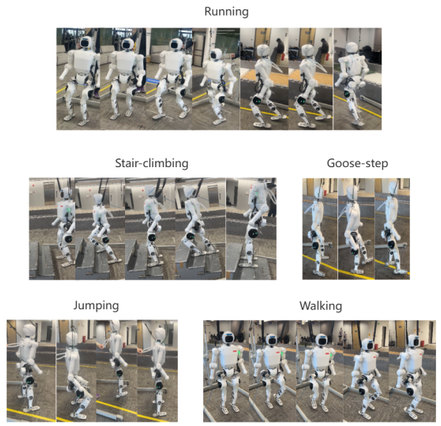


Fig. 2. Representative real-robot image sequences for the five learned gaits: walking, stair climbing, goose-stepping, jumping, and running.

Figure 3 compares goose-stepping with AMP and jumping without AMP as representative cases. Under AMP, goose-stepping exhibits a rapid increase in total reward, low training error, and a prompt decay in failure rate. The main contribution of AMP in this setting is not merely a higher terminal return, but the reinforcement of periodic structure and style consistency, which is especially important for a gait with explicit kinematic features such as high leg lift and straight-knee swing. For running, a strong AMP prior would not necessarily render the policy incapable of locomotion, but it would likely increase behavioral conservatism and bias the solution toward a more strongly regularized motion pattern. In that regime, the learned behavior would be more likely to resemble a smooth fast walk or a weak running pattern, rather than a genuinely explosive gait with clear aerial phases, strong forward propulsion, and large leg excursion. This observation suggests that, for highly dynamic locomotion, an excessively strong motion prior may improve motion regularity while simultaneously compressing the policy's access to high-amplitude dynamic solutions. By contrast, jumping

表 II 五种步态任务的主要训练配置。

Gait	# 环境	观测维度	动作维度	模拟步长	控制周期	每个环境的步数 _	最大迭代次数	*AMP*组
行走	4096	1050	12	0.005 秒	0.02 秒	24	40001	AMP
正步走	4096	1050	12	0.005 s	0.02 s	24	20001	AMP
爬楼梯	4096	1050	12	0.005 s	0.02 s	24	40001	AMP
跑步	2048	1050	12	0.005 s	0.02 s	24	40001	无AMP
跳跃	2048	1050	12	0.005 s	0.02 秒	24	11000	无 AMP

反复维持期望的步态，而高跌倒率通常表明对初始条件、节奏控制或动态恢复的敏感性。训练误差表征了优化的平滑度：较低误差通常对应更稳定的参数更新，而持续较大的误差则表明更强的振荡和更高的不确定性。姿态稳定性衡量全身控制质量。更高的姿态稳定性对应更小的躯干振荡、更平滑的横滚-俯仰调节以及更一致的相位转换。以下使用这些指标来比较整体步态性能以及AMP与非AMP训练之间的区别。

C. 结果与分析

1) 五种步态的总体性能：在统一的策略结构、共享的观测设计以及共同的奖励组织下，所提出的框架学习和部署了五种运动模式：行走、齐步、爬楼梯、奔跑和跳跃。这些任务涵盖了从低动态周期性运动到高动态爆发性运动的整个范围，从而测试了统一强化学习公式的可扩展性，使其能够跨越显著不同的步态类别。

从定性角度来看，行走呈现出规律的向前迈步模式；齐步表现出预期的抬高腿和直腿摆动；爬楼梯在上升和下降过程中都保持足够的脚部离地间隙和身体姿势；奔跑产生更短的支撑时间和更强的向前推进力；跳跃则实现了从蹲伏准备和快速起跳到空中稳定和着陆恢复的完整运动链。图2展示了五种步态的代表性真实机器人图像序列。这些结果表明，统一框架不仅能够保持平衡，还能够生成具有明显不同动态特性和风格特征的运动模式。

表III总结了五种步态的定量性能。所报告的指标对应于关节跟踪质量、收敛速度、任务可靠性、训练稳定性和姿态稳定性。由于奖励组成和任务难度在不同步态中有所不同，这些值旨在表征在统一框架下的训练和执行行为，而不是作为跨任务的严格归一化物理量。

2) AMP对不同步态的影响：本小节考察了AMP在步态类别间的操作边界。通常，AMP对具有清晰周期结构和以稳定性为先目标的步态最为有效，而它则

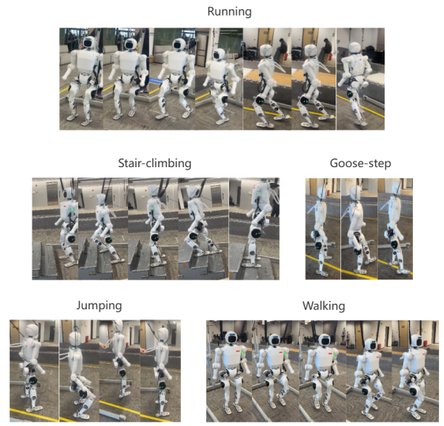


图2. 五种学习到的步态的代表性真实机器人图像序列：行走、爬楼梯、齐步走、跳跃和跑步。

对于高度动态和强烈爆发性的运动，它可能会变得限制性。行走、正步和爬楼梯属于前者：它们需要稳定的站立-摆动过渡、一致的关节协调和低姿态波动。对于这类任务，运动先验可以减少无效探索，通常能产生更低的跟踪误差、更平滑的奖励增长和更快的收敛速度。

图3将行进间踏步与带AMP的行进间踏步以及不带AMP的跳跃作为典型案例进行了比较。在AMP条件下，行进间踏步表现出总奖励的快速增加、训练误差低以及失败率的快速衰减。在此设置中，AMP的主要贡献并不仅仅是更高的终端回报，而是对周期性结构和风格一致性的强化，这对于具有明确运动学特征的步态（如高抬腿和直腿摆动）尤为重要。对于跑步，强大的AMP先验并不会必然使策略丧失运动能力，但它可能会增加行为保守性，使解决方案倾向于更强烈的正则化运动模式。在这种情况下，学习到的行为更可能类似于平滑快走或弱跑模式，而不是具有明显腾空阶段、强向前推进和较大腿部摆动幅度的真正爆发性步态。这一观察表明，对于高度动态的运动，过强的运动先验可能会在提高运动规律性的同时压缩策略对高幅动态解决方案的访问能力。相比之下，跳跃

TABLE III

QUANTITATIVE SUMMARY OF THE FIVE GAIT TASKS.

Gait	Joint-Tracking Metric	Convergence Steps	Estimated Success Rate	Training Error	Posture-Stability Metric	Motion Characteristic
Walking	0.40	2500	95%	10.0	2.3	Periodic and stable forward locomotion
Goose-stepping	0.94	2500	97%	5.4	1.2	High leg lift and straight-knee swing
Stair climbing	1.00	5200	85%	1.0	0.95	Stair ascent/descent with recovery
Running	0.50	3000	96%	4.8	1.3	Short stance and strong propulsion
Jumping	15.5	3000	83%	6.0	8.6	Two-foot takeoff and landing recovery

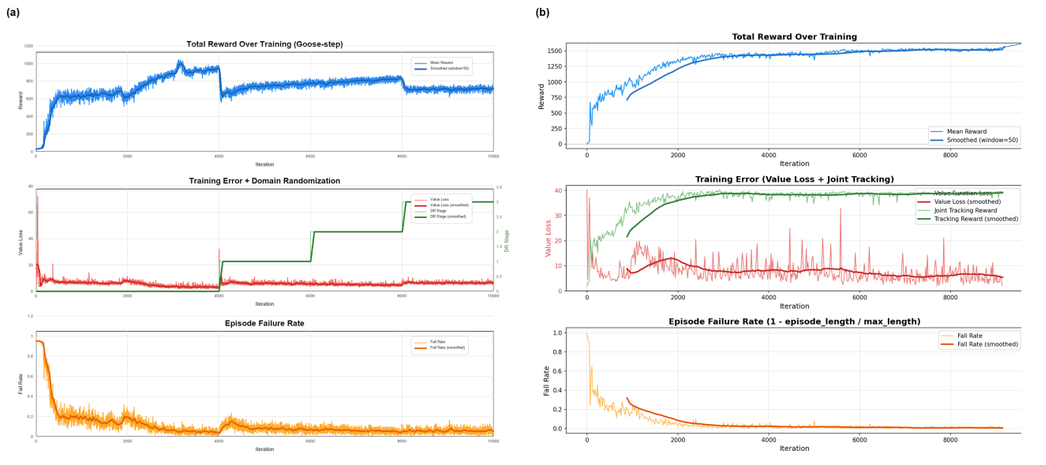


Fig. 3. Training curves used in the AMP-versus-no-AMP comparison. (a) Goose-stepping with AMP. (b) Jumping without AMP. Each panel reports total reward, training error or auxiliary metric, and failure rate.

depends on large motion amplitude, vertical impulse, and aerial freedom. Under the non-AMP setting, the task still attains sustained reward growth and low failure rate while the joint-tracking signal improves steadily. This indicates that task rewards alone are sufficient to drive effective takeoff, flight, and landing recovery when expressive exploration is preserved. These results support the view that AMP should be treated as a gait-dependent mechanism rather than a uniformly active module.

3) Real-Robot Robustness and Practical Performance: Beyond single demonstrations, the engineering value of the method is reflected in sustained execution on the physical robot. We therefore assess real-robot robustness from three perspectives: long-horizon continuity, consistency across repeated trials, and recovery under mild external disturbances or start–stop transitions.

The robustness criteria vary across gaits. For walking and goose-stepping, the emphasis lies in rhythm consistency and controlled torso oscillation; for stair climbing, in foot clearance, step placement, and posture recovery after stair transitions; for running, in stability during rapid short-stance switching; and for jumping, in synchronized takeoff, aerial posture control, landing absorption, and return to the next motion cycle. In particular, the jumping reference is explicitly divided into squat, takeoff, flight, landing, and standing-recovery phases, so robustness depends not only on leaving the ground, but also on the quality of aerial motion and landing recovery.

Taken together, the real-robot deployments show that the learned policies are not limited to visually plausible motions in simulation, but exhibit the continuity and repeatability required for physical execution. This supports the practical viability of the proposed unified framework for multi-gait humanoid locomotion.

IV. CONCLUSION

This paper presented a unified reinforcement learning framework for multi-gait humanoid locomotion together with a selective AMP strategy. Within a common policy architecture and reward formulation, the proposed method learned five representative gaits, namely walking, goose-stepping, stair climbing, running, and jumping, and transferred them to a physical humanoid platform. The results indicate that motion priors are beneficial for periodic, stability-oriented gaits, but should be weakened or removed for highly dynamic behaviors that depend on expressive motion generation. More importantly, the proposed framework provides a practical and reproducible baseline for multi-gait learning on real humanoid hardware by combining a unified RL formulation, gait-dependent prior selection, and real-robot validation. Overall, this study suggests that practical humanoid mobility will depend not only on stronger policies, but also on principled methods for organizing diverse behaviors within a shared learning framework.

表 III 五种步态任务的定量总结。

Gait	关节跟踪指标	收敛步数	估计成功率	训练误差	姿态稳定性指标	运动特征
行走	0.40	2500	95%	10.0	2.3	周期性稳定的向前移动
踏步	0.94	2500	97%	5.4	1.2	高抬腿和伸直膝关节摆动
爬楼梯	1.00	5200	85%	1.0	0.95	上下楼梯时的恢复
跑步	0.50	3000	96%	4.8	1.3	短支撑和强力推进
跳跃	15.5	3000	83%	6.0	8.6	双脚起跳和落地恢复

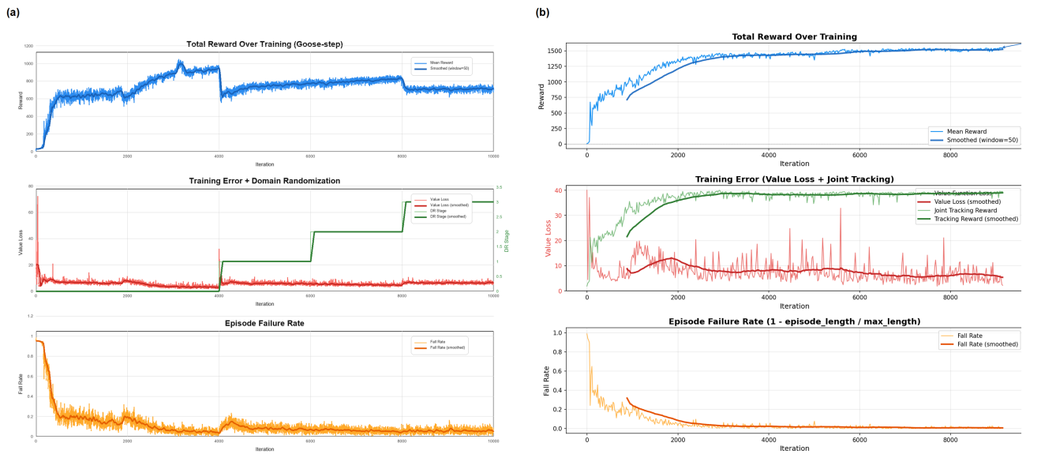


图3. AMP与无AMP对比中使用的训练曲线。(a) 带AMP的踏步。(b) 无AMP的跳跃。每个面板报告总奖励、训练误差或辅助指标以及失败率。

则依赖于较大的运动幅度、垂直冲量和腾空自由度。在非AMP设置下，任务仍然实现了持续的奖励增长和低失败率，同时关节跟踪信号稳步改善。这表明，当保持表达性探索时，仅靠任务奖励就足以驱动有效的起跳、飞行和着陆恢复。这些结果支持将AMP视为一种与步态相关的机制，而不是一个始终活跃的模块的观点。

3) 实际机器人鲁棒性与实用性能：超越单一演示，该方法的价值体现在物理机器人上的持续执行。因此，我们从三个角度评估实际机器人的鲁棒性：长时程连续性、重复试验的一致性，以及在轻微外部干扰或启停转换下的恢复能力。

不同步态的鲁棒性标准有所不同。对于行走和正步，重点在于节奏一致性和受控的躯干摆动；对于爬楼梯，重点在于脚部离地高度、步点位置以及楼梯转换后的姿态恢复；对于跑步，重点在于快速短支撑切换时的稳定性；而对于跳跃，重点在于同步起跳、空中姿态控制、落地缓冲以及返回下一个运动周期。特别是跳跃参考被明确分为下蹲、起跳、飞行、落地和站立恢复阶段，因此鲁棒性不仅取决于离地，还取决于空中动作的质量和落地恢复。

综合来看，实际机器人部署表明，所学习的策略不仅限于模拟中的视觉合理运动，而且表现出物理执行所需的连续性和可重复性。这支持了所提出的统一框架在多足人形机器人运动中的实用可行性。

IV. 结论

本文提出了一种统一的强化学习框架，用于多步态人形机器人运动，并结合了选择性AMP策略。在通用的策略架构和奖励公式下，所提出的方法学习了五种代表性步态，即行走、齐步、爬楼梯、跑步和跳跃，并将它们迁移到物理人形机器人平台上。结果表明，运动先验对于周期性、以稳定性为导向的步态有益，但对于依赖表现力运动生成的、高度动态的行为则应减弱或移除。更重要的是，该框架通过结合统一的RL公式、步态相关先验选择和真实机器人验证，为真实人形硬件上的多步态学习提供了一个实用且可复现的基线。总体而言，本研究表明，实用的人形机器人移动不仅取决于更强的策略，还取决于在共享学习框架内组织多样化行为的原则性方法。

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